NARAYANA IIT - JEE ACADEMY - INDIA

Sec: SR.IIT_*CO-SC(MODEL-A& APEX) Date:24-12-24

Time: 3HRS GTM-3 Max. Marks: 300

KEYSHEET

PHYSICS

01	02	03	04	05	06	07	08	09	10
A	A	A	B	B	A	B	D	B	A
11	12	13	14	15	16	17	18	19	20
B	A	C	C	A	B	A	C	B	C
21	22	23	24	25					
300	9	7	5	50					

CHEMISTRY

26	27	28	29	30	31	32	33	34	35
C	B	D	B	C	B	D	A	C	B
36	37	38	39	40	41	42	43	44	45
D	C	A	C	C	A	C	D	C	D
46	47	48	49	50					
6	5	24	36	5					

MATHEMATICS

51	52	53	54	55	56	57	58	59	60
C	B	C	C	B	B	D	D	A	C
61	62	63	64	65	66	67	68	69	70
D	C	D	A	C	A	D	C	D	B
71	72	73	74	75					
10	7	339	6	81					

APEX JEE-MAIN 2025

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SOLUTIONS

PHYSICS

01. $(N+m)VSD=(N)MSD$

$\therefore \quad 1VSD=\left(\frac{N}{N+m}\right)MSD$

$LC=1MSD-1VSD$

$=1MSD-\left(\frac{N}{N+m}\right)MSD$

$=\left(\frac{1}{1+N/m}\right)MSD$

For minimum value of least count m should be minimum or 1.

02. Area under $a-t$ graph gives the change in velocity $(dv=adt)$

$\therefore \quad v_f-v_i=\frac{1}{2}\times 2\times 4=4\,m/s \qquad a=-\omega^2 x$

$\therefore \quad v_f=v_i+4=2+4=6\,m/s$

$\omega^2=\tan 45^\circ=1$

03. $KE_{trans}=\frac{3}{2}nRT=\frac{3}{2}\times 2\times 8.31\times 293=7.3\times 10^3 J$

04. Acceleration

i.e.,

or $\frac{2\pi}{T}=1$ or $T=2\pi s$

05. $F_A=F_B=(P_0+\rho gh)$ (base area)

06. Let I be the desired length. Chain starts moving when net pulling force = maximum force of friction

or $g.\frac{m}{L}.I=\mu\frac{m}{L}(L-1).g$

(m=mass of complete length of chain)

or $I=\left(\frac{\mu}{\mu+1}\right)L$

07. Let v_b is velocity of bullet, v_p is velocity of police van and v_t is velocity of thief's car. Then, speed of the bullet with respect to the thief's car,

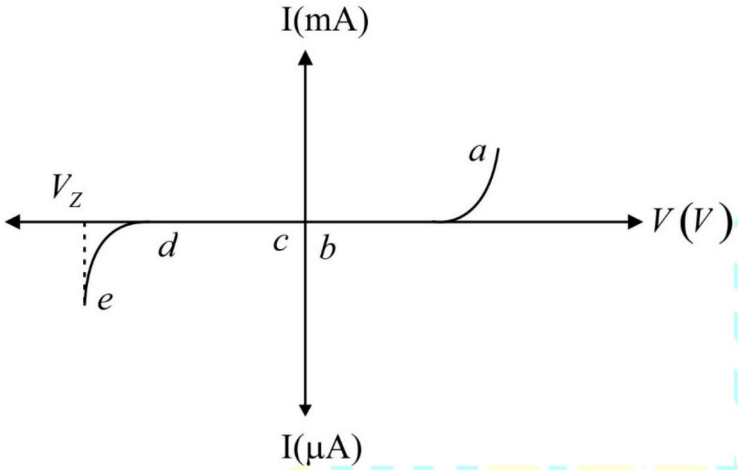
$v_{bp}=v_b-v_p$

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$$v_b = v_{bp} + v_p = 150 \times \frac{18}{5} kmh^{-1} + 30 kmh^{-1} = 570 kmh^{-1}$$

$$v_{bt} = v_b + v_t = 570 kmh^{-1} - 190 kmh^{-1} = 380 kmh^{-1}$$

08.



If the reverse bias voltage is greater than the V_Z, then there is breakdown condition. In breakdown region, i.e., $V_i > V_z$ and for a long range of load (R_L), the voltage is constant.

If the voltage of reverse bias is greater than the zener voltage a breakdown condition is obtained.

$\therefore$ in the given graph in the part de, current is increasing but voltage in nearly same

09. The excess pressure of soap bubble.

$$p = \frac{4T}{R}$$

$$or\ h\rho g = \frac{4T}{R} \qquad (\because p = h\rho g)$$

$$\Rightarrow T = \frac{Rh\rho g}{4} = \frac{1 \times 10^{-2} \times 2 \times 10^{-3} \times 0.8 \times 10^3 \times 9.8}{4}$$

$$= 0.0392\ Nm^{-1}$$

10. As, force, $F = \frac{e^2}{4\pi\varepsilon_0 r^2} \Rightarrow \frac{e^2}{4\pi\varepsilon_{0-}} = r^2.F$

Putting dimensions of r and F, we get.

$$\Rightarrow \left[\frac{e^2}{4\pi\varepsilon_0}\right] = \left[ML^3T^{-2}\right] \ldots\ldots\ldots\ldots\ldots (I)$$

Also, force, $F = \frac{Gm^2}{r^2} \Rightarrow [G] = \left[M^{-1}L^3T^{-2}\right] \ldots\ldots (ii)$

$$and \left[\frac{1}{c^2}\right] = \frac{1}{\left[L^2T^{-2}\right]} = \left[L^{-2}T^2\right] \ldots\ldots (iii)$$

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Now, checking options wise

$$=\frac{1}{c^2}\left(\frac{Ge^2}{4\pi\varepsilon_0}\right)^{1/2}=\left[L^{02}T^2\right]\left[L^6T^{-4}I^{1/2}=[L]\right]$$

11. The energy of artificial satellite at the surface of the earth,

$$E_1=-\frac{GMm}{R}$$

When the satellite is intended to move in a circular orbit of radius 7R, then energy of artificial satellite,

From conservation of energy

$$(KE+PE)_{surface}=(KE+PE)_{orbit}$$

$$-\frac{GMm}{R}+KE_{\min}=-\frac{GMm}{2\times 7R}$$

$$KE_{\min}=\frac{13}{14}\left(\frac{GMm}{R}\right)$$

12. $T-mg=ma$

$\therefore \quad T=m(g+a)$

$$W_T=TS=T\left(\frac{1}{2}at^2\right)=\frac{m}{2}(g+a)at^2$$

13. In adiabatic process

$PV^{\gamma}=$ constant

$$\therefore P\left(\frac{m}{p}\right)^{\gamma}=\text{constant}\left(\therefore V=\frac{m}{\rho}\right)$$

As mass is constant

$\therefore P\propto\rho^{\gamma}$

If P_i and P_f be the initial and final pressure of the gas and ρ_i and ρ_f be the initial and final density of the gas. Then

$$\frac{P_f}{P_i}=\left(\frac{\rho_f}{\rho_i}\right)^{\gamma}=(32)^{7/5}$$

$$\Rightarrow n\frac{P_i}{P_f}=\left(2^5\right)^{7/5}=2^7\Rightarrow n=2^7=128.$$

14. For first law of the thermodynamics

$$\Delta Q=\Delta Q-\Delta W$$

Here $\Delta Q=mL=L$

and $\Delta W=P_0(V_t-V_i)=P_0(V_2-V_1)$

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$\therefore \quad \Delta U = L - P_0(V_2 - V_1)$

15. Torque about the origin $= \vec{\tau} = \vec{r} \times \vec{F}$

$= rF\sin\theta \Rightarrow 2.5 = 1 \times 5\sin\theta$

$\sin\theta = 0.5 = \frac{1}{2}$

$\Rightarrow \theta = \frac{\pi}{6}$

16. $I_x = I_y = 2\left[\frac{ml^2}{3}\sin^2 45^\circ\right] = \frac{ml^2}{3}$

$I_z = 2\left[\frac{ml^2}{3}\right] = \frac{2ml^2}{3}$

17. $\frac{3l}{k'} = \frac{1}{k/2} + \frac{1}{5k} + \frac{1}{k}$

18. According to Lami's theorem, $\frac{T_1}{\sin 120^0} = \frac{T_2}{\sin 120^0} = \frac{10}{\sin 120^0}$

$\Rightarrow T_1 = 10N, T_2 = 10N$

Thus, the tension in section BC and BF are 10 N and 10 N, respectively,

19. $v = \sqrt{2gh}, t = \sqrt{\frac{2h}{g}}$ and $R = 2\sqrt{h(H-h)}$

20. Here C =A, B and $D = \overline{A}.B$

$E = C + D = (A.B) + (\overline{A}.B)$

Explanation. The truth table of this arrangement of gates can be given below:

A	B	$\overline{A}$	$C = A.B$	$D = \overline{A}.B$	$E = (C + D)$
0	0	1	0	0	0
0	1	1	0	1	1
1	0	0	0	0	0
1	1	0	1	0	1

21. From work, energy theorem, $W = \Delta KE$

$= K_f - K_i = \frac{1}{2}m\left(V_f^2 - V_i^2\right) = \frac{1}{2}(2)(400 - 100) = 300J$

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22. Let σ be the mass per unit area of uniform plate.

$\therefore$ Mass of the whole disc $= \sigma \times \pi R^2$

Mass of the portion removed $= \sigma \times \pi r^2$

$R = 28$ cm; $r = 21$ cm; $OP = 7$ cm Position of centre of mass

$$x = \frac{m_1 x_1 - m_2 x_2}{m_1 - m_2}$$

$$= \frac{\sigma \times \pi R^2(0) - \sigma \times \pi r^2 \times 7}{\sigma\pi R^2 - \sigma\pi r^2}$$

$$= \frac{-(21)^2 \times 7}{(28)^2 - (21)^2} = -\frac{21 \times 21 \times 7}{7 \times 49} = -9 \text{ cm}$$

i.e., centre of mass lies at a distance of 9 cm from the origin towards left

23. $a_c = \frac{a_A + a_B}{2}$

24. According to given condition,

$A + B\hat{i} = R\hat{j}$ (where, R is the resultant vector)

Also, R = 2B

$\therefore A + B\hat{i} = 2B\hat{j}$ or $A = 2B\hat{j} - B\hat{i}$

$\therefore A^2 = 4B^2 + B^2 = 5B^2$

Here, A = 8

$\Rightarrow B = \sqrt{\frac{64}{5}} = \frac{8}{\sqrt{5}} = \frac{8}{\sqrt{x}} \Rightarrow x = 5$

25. Given, $v = a x^{3/2}$

$m = 0.5\,kg$, $a = 5m^{-1/2}s^{-1}$, work done, W = ?

Acceleration, $a_0 = \frac{dv}{dt} = v\frac{dv}{dx} = ax^{3/2}\frac{d}{dx}\left(ax^{3/2}\right)$

$= ax^{3/2} a \times \frac{3}{2} \times x^{1/2} = \frac{3}{2}a^2x^2$

Now, force $= ma_0 = m\frac{3}{2}a^2x^2$

Work done, $W = \int_{x=0}^{x=2} F dx = \int_0^2 \frac{3}{2} ma^2x^2 dx = \frac{3}{2}ma^2 \times \left[\frac{x^3}{3}\right]_0^2$

CHEMISTRY

26. The balanced equation is, $6I^- + ClO_3^- + 6H_2SO_4 \rightarrow Cl^- + 6HSO_4^- + 3I_2 + 3H_2O$

27. Let 'x' is the % by mass of carbon. Given the weight gained per 100kg = 4kg

i.e., $\frac{x}{12} \times 14 - x = 4$

$\Rightarrow \quad x = 24$

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28. Conceptual

29. A reaction with ΔH=-ve and ΔS = +ve is spontaneous at all temperatures.

30. $H_2C_2O_4(s) + \frac{1}{2} O_2(g) \rightarrow H_2O(l) + 2CO_2(g)$;

$\Delta n_g = 3/2$ per mole

$\Delta U_c = -\frac{0.312 \times 8.75}{1} \times 90 = -245.7$ kJ/mol

$\Delta H = \Delta U + \Delta n_g RT$

$= -245.7 + \frac{3}{2} \times \frac{8.314 \times 300}{1000} = -241.947$ kJ/mol.

31. $N_2 + 3H_2 \rightleftharpoons 2NH_3$

t = 0 1 mole 2 mole 0

t = eq 1–x 2–3x 2x = 0.8

x = 0.4

mole of N_2 = 0.6

mole of H_2 = 0.8

32. A reaction with Δn = +ve has maximum value and -ve has minimum value.

33. Equilibrium proceeds in the direction from strong (Acid/Base) to weak (Acid/Base).

I : $SA + SB \rightarrow WA + WB$

Here, acidic strength is compared among the 2 acids in the reaction & basic strength is compared among 2 bases in that reaction only.

II : $WA + WB \rightarrow SA + SB$

34. The order of atomic radius is; Be > B > C > N > O.

35. Reactivity of alkali metals increases down the group and halogens decrease down the group.

36. (vi) is non aromatic and (viii) is antiaromatic

37. As pH decreases, $[H^+]$ increases. So, CN^- will combine with H^+ forming HCN making the solubility equilibrium shift in forward direction.

38. Stability $\propto$ Number of hyperconjugative H-atoms.

39. Conceptual

40. Lead acetate gives black precipitate of PbS with sulphide salts.

41. Conceptual

42. Neopentane has a compact structure with two side chains. It gives scope for close

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packing and intermolecular attractions are strong. It has higher M.P than other.

43. As no intermediate is formed during E_2 no rearrangement takes place. Other statements are correct

44. The directing effect of halogens is due to mesomeric effect and deactivation is because of electronegativity

45. Liquids differing more than 30^0 C in boiling point are separated by fractional distillation.

46. $CO_2 + 2KOH \rightarrow K_2CO_3 + H_2O$

No. of moles of CO_2 formed $= \frac{300 \times 2 \times 10^{-3}}{2} = \frac{0.6}{2} = 0.3$

$$C_nH_{2n+2} + \left(\frac{3n+1}{2}\right)O_2 \longrightarrow nCO_2 + (n+1)H_2O$$

1 mole $\rightarrow$ n moles

$\frac{4.3}{(12n+2n+2)}$ mole $\rightarrow$?

$\frac{4.3}{14n+2}.n = 0.3$

$4.3n = 4.2n + 0.6$

$0.1n = 0.6$

$n = 6$

47. In compounds i, ii, iv, vii, viii the central atom has two lone paires.

48. $\Delta E = 0 \quad \Rightarrow \quad 20 \times 4.2 - 60 + W = 0$

$W = -24$ J.

49. Conceptual

50. A) HBr D) $Hg(OAc)_2/H_2O$, $NaBH_4/OH^-$ E) aq.H_2SO_4 F) H_2 / Ni H) NBS/hv

MATHEMATICS

51. $[x]^2 - 3[x] - 10 > 0 \Rightarrow [x]^2 - 5[x] + 2[x] - 10 > 0$

$[x] < -2$ or $[x] > 5 \Rightarrow (-\infty, -2) \cup [6, \infty)$

52. $\cos^2 10° - \cos 10° \cos 50° + \cos^2 50°$

$$= \left(\frac{1+\cos 20°}{2}\right) + \left(\frac{1+\cos 100°}{2}\right) - \frac{1}{2}(2\cos 10° \cos 50°)$$

$$= 1 + \frac{1}{2}(\cos 20° + \cos 100°) - \frac{1}{2}[\cos 60° + \cos 40°]$$

$$= \left(1 - \frac{1}{4}\right) + \frac{1}{2}[\cos 20° + \cos 100° - \cos 40°]$$

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$=\frac{3}{4}+\frac{1}{2}\left[2\cos 60^\circ \times \cos 40^\circ - \cos 40^\circ\right]=\frac{3}{4}$

53. Given $3\cos^4\theta - 5\cos^2\theta - 2\sin^6\theta + 2 = 0$

$\Rightarrow 3\cos^4\theta - 3\cos^2\theta - 2\cos^2\theta - 2\sin^6\theta + 2 = 0$

$(\because$ as $2 - 2\cos^2\theta = 2\sin^2\theta)$

$\Rightarrow 3\cos^4\theta - 3\cos^2\theta + 2\sin^2\theta - 2\sin^6\theta = 0$

$\Rightarrow 3\cos^2\theta\left(\cos^2\theta - 1\right) + 2\sin^2\theta\left(1 - \sin^4\theta\right) = 0$

$\Rightarrow -3\cos^2\theta\sin^2\theta + 2\sin^2\theta\left(1 + \sin^2\theta\right)\cos^2\theta = 0$

$\Rightarrow \sin^2\theta\cos^2\theta\left(2\sin^2\theta - 1\right) = 0$

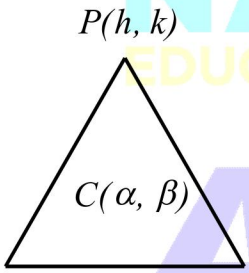
Case I: $\sin^2\theta = 0 \rightarrow 3$ solution; $\theta = (0, \pi, 2\pi)$

Case II: $\cos^2\theta = 0 \rightarrow 2$ solution; $\theta = \left\{\frac{\pi}{2}, \frac{3\pi}{2}\right\}$

Case III: $\sin^2\theta = \frac{1}{2} \rightarrow 4$ solution; $\theta = \left\{\frac{\pi}{4}, \frac{3\pi}{4}, \frac{5\pi}{4}, \frac{7x}{4}\right\}$

No. of solution= 9

54.



Let centroid C be (α, β)

We have $\alpha = \frac{1+3+h}{3} \Rightarrow h = 3\alpha - 4$

$\beta = \frac{4-2+k}{3} \Rightarrow k = 3\beta - 2$

But $P(h,k)$ lies on $2x - 3y + 4 = 0$

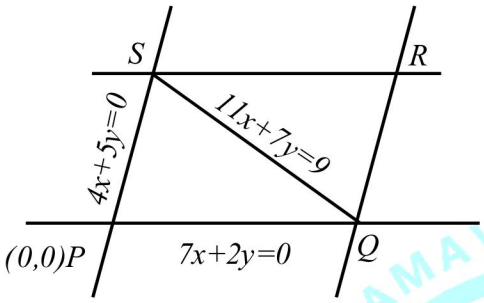
$\Rightarrow 2(3\alpha - 4) - 3(3\beta - 2) + 4 = 0$

$\Rightarrow 6\alpha - 9\beta + 2 = 0$

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Locus: $6x-9y+2=0 \Rightarrow y=\frac{6}{9}x+\frac{2}{9} \therefore$ its slope $=\frac{6}{9}=\frac{2}{3}$

55. Since both the lines pass through origin. Then



Point S will be point of intersection of $4x+5y=0$ and $11x+7y=9$

So, coordinates of point $S=\left(\frac{5}{3},-\frac{4}{3}\right)$

Similarly, $Q=\left(-\frac{2}{3},\frac{7}{3}\right)$

Since, diagonals of parallelogram intersect at middle, then the middle point of SQ is

$$\left(\frac{\frac{5}{3}-\frac{2}{3}}{2},\frac{\frac{-4}{3}+\frac{7}{3}}{2}\right)=\left(\frac{1}{2},\frac{1}{2}\right)$$

So, equation of diagonal PR is,

$$(y-0)=\frac{\frac{1}{2}-0}{\frac{1}{2}-0}(x-0) \Rightarrow y=x$$

Hence, diagonal PR passes through the point $(2,2)$

56. $\lim\limits_{t\to 0}\left(1^{\frac{1}{\sin^2 t}}+2^{\frac{1}{\sin^2 t}}+\ldots+n^{\frac{1}{\sin^2 t}}\right)^{\sin^2 t}$

$$=\lim_{t\to 0}\left(1^{\operatorname{cosec}^2 t}+2^{\operatorname{cosec}^2 t}+\ldots.+n^{\operatorname{cosec}^2 t}\right)^{\sin^2 t}$$

$$=\lim_{t\to 0} n\left(\left(\frac{1}{n}\right)^{\operatorname{cosec}^2 t}+\left(\frac{2}{n}\right)^{\operatorname{cosec}^2 t}+\ldots.+1\right)^{\sin^2 t}=n(0+0+0+\ldots.+1)=n$$

57. The given limit is,

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$$\lim_{x\to 0}\frac{\alpha x\left(1+x+\frac{x^2}{2}\right)-\beta\left(x-\frac{x^2}{2}+\frac{x^3}{3}\right)+\gamma x^2(1-x)}{x\sin^2 x}\times\frac{x^2}{x^2}$$

$$\lim_{x\to 0}\frac{x(\alpha-\beta)+x^2\left(\alpha+\frac{\beta}{2}+\gamma\right)+x^3\left(\frac{\alpha}{2}-\frac{\beta}{3}-\gamma\right)}{x^3}=10$$

$\because$ limit exists, then

$\alpha-\beta=0$ …(i)

$\alpha+\frac{\beta}{2}+\gamma=0$ …(ii)

and $\frac{\alpha}{2}-\frac{\beta}{3}-\gamma=10$ …(iii)

Then from (i) and (ii); $\beta=\alpha,\gamma=-3\frac{\alpha}{2}$

Now, put $\beta-\alpha$ and $\gamma=\frac{-2\alpha}{2}$ in eq^n (iii)

$$\frac{\alpha}{2}-\frac{\alpha}{3}+\frac{3\alpha}{2}=10\Rightarrow\frac{\alpha+9\alpha}{6}=10\Rightarrow\alpha=6$$

$\therefore\ \alpha=6,\beta=6,\gamma=-9;$ so, $2\alpha+3\beta-\gamma=39$

58. $y=\frac{8^{2x}-8^{-2x}}{8^{2x}+8^{-2x}}\Rightarrow\frac{1+y}{1-y}=\frac{8^{2x}}{8^{-2x}}\Rightarrow 8^{4x}=\frac{1+y}{1-y}$

$$\Rightarrow 4x=\log_8\left(\frac{1+y}{1-y}\right)\Rightarrow x=\frac{1}{4}\log_8\left(\frac{1+y}{1-y}\right)$$

$$\therefore f^{-1}(x)=\frac{1}{4}\log_8\left(\frac{1+x}{1-x}\right)$$

59. Given, $A^T=A,B^T=-B,C^T=-C$

Let $M=A^{13}B^{26}-B^{26}A^{13}$

Then, $M^T=\left(A^{13}B^{26}-B^{26}A^{13}\right)^T$

By using property $(AB)^T=B^TA^T$

$=\left(A^{13}B^{26}\right)^T-\left(B^{26}A^{13}\right)^T=B^{26}A^{13}-A^{13}B^{26}$

Hence, M is skew symmetric

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Let, $N = A^{26}C^{13} - C^{13}A^{26}$

Then, $N^T = \left(A^{26}C^{13}\right)^T - \left(C^{13}A^{26}\right)^T = -(C)^{13}(A)^{26} + A^{26}C^{13} = N$

Hence, N is symmetric, $\therefore$ Only S2 is true.

60. Given, $P^2 = I - P$,

$P^\alpha + P^\beta = \gamma I - 29P, P^\alpha - P^\beta = \delta I - 13P$

Now, $P^4 = (I-P)^2 = I - 2P + P^2 = 2I - 3P$

$P^6 = (2I - 3P)(I - P) = 5I - 8P$

$P^8 = (2I - 3P)^2 = 4I - 12P + 9(I - P) = 13I - 21P$

$P^8 + P^6 = 18I - 29P$ and $P^8 - P^6 == 81 - 13P$

So, $\alpha = 8; \beta = 6; \gamma = 18, \delta = 8$

$\Rightarrow \alpha + \beta + \gamma - \delta = 8 + 6 + 18 - 8 = 24$

61. Given system of equations are $x + y + z = 6$

$2x + 5y + \alpha z = \beta$

$x + 2y + 2z = 14$

From the given equations.

$x + y = 6 - z$ …(i)

$x + 2y = 14 - 3z$ …(ii)

Subtract (i) from (ii),

$\Rightarrow y = 8 - 2z$, then $x = z - 2$

Now, put the values x & y

In eq. $2x + 5y + \alpha z = \beta$

$2(z-2) + 5(8 - 2z) + \alpha z = \beta$

$(\alpha - 8)z = \beta - 36$

For having infinite solutions

$\alpha - 8 = 0 \,\&\, \beta - 36 = 0 \Rightarrow \alpha = 8, \beta = 36$

Required sum $= 2\alpha + \beta = 52$

62. Since $x - [x] \geq 0 \forall \Rightarrow x \in R$

Need to check at doubtful points

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$$f(x)=\begin{cases}2+\sqrt{x+2}, & -2<x<-1\\ 1+\sqrt{x+1}, & -1\le x<0\\ \sqrt{x}, & 0\le x<1\end{cases}$$

At $x=-1\Rightarrow f\left(-1^{+}\right)=1+0=1$

$\Rightarrow f\left(-1^{-}\right)=2+1=3$

At $x=0\Rightarrow f\left(0^{+}\right)=0+0=0$

$\Rightarrow f\left(0^{-}\right)=1+1=2$

Discontinuous at two points $x=\{-1,0\}$

63. $f(x)$ is continuous at $x=0$

$\Rightarrow \lim\limits_{x\to 0} f(x)=f(0)=3$

$$\lim_{x\to 0^{+}}\frac{\sqrt{ax+b^2x^2}-\sqrt{ax}}{b\sqrt{ax}\sqrt{x}}=3$$

By Rationalization we get

$$\lim_{x\to 0^{+}}\frac{ax+b^2x^2-ax}{b\sqrt{a}x^{3/2}\left(\sqrt{ax+b^2x^2}+\sqrt{ax}\right)}$$

$$\lim_{x\to 0^{+}}\frac{b^2}{b\sqrt{a}\left(\sqrt{a+b^2x}+\sqrt{a}\right)}$$

$$\frac{b}{\sqrt{a}.2\sqrt{a}}\Rightarrow\frac{b}{2a}=3\Rightarrow\frac{b}{a}=6$$

64. $f(x+y)=f(x)+f(y)+xy^2+x^2y$

Differentiate w.r.t. x: [taking y as constant]

$f'(x+y)=f'(x)+0+y^2+2xy$

Put $y=-x$

$f'(0)=f'(x)+x^2-2x^2$ …(i)

$\because \lim\limits_{x\to 0}\frac{f(x)}{x}=1\Rightarrow f(0)=0$

$\therefore f'(0)=1$ …(ii)

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From equations (i) and (ii),

$f'(x)=\left(x^2+1\right)\Rightarrow f'(3)=10$

65. It is given that

$x=2\sin\theta-\sin 2\theta \quad \ldots(i)$

$y=2\cos\theta-\cos 2\theta \quad \ldots(ii)$

$\frac{dx}{d\theta}=2\cos\theta-2\cos 2\theta$

Differentiating (ii) w.r.t θ; we get

$\frac{dy}{d\theta}=-2\sin\theta+2\sin 2\theta \ \therefore \frac{dy}{dx}=\frac{\sin 2\theta-\sin\theta}{\cos\theta-\cos 2\theta}$

$=\frac{2\sin\frac{\theta}{2}.\cos\frac{3\theta}{2}}{2\sin\frac{\theta}{2}.\sin\frac{3\theta}{2}}=\cot\frac{3\theta}{2}$

Again, differentiating eqn. (iii), we get

$\frac{d^2y}{dx^2}=\frac{-3}{2}\cos ec^2\frac{3\theta}{2}.\frac{d\theta}{dx}$

$\frac{d^2y}{dx^2}=\frac{\frac{-3}{2}\cos ec^2\frac{3\theta}{2}}{2\left(\cos\theta-\cos 2\theta\right)}$

$\frac{d^2y}{dx^2\left(\theta=\pi\right)}=\frac{3}{4(-1-1)}=\frac{3}{8}$

66. Given $y^2=18x\Rightarrow 2y\frac{dy}{dx}=18\Rightarrow\frac{dy}{dx}=\frac{9}{y}$

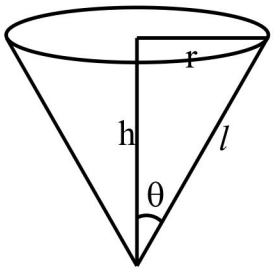
$ATQ\ \frac{dy}{dt}=\frac{2dx}{dt}\Rightarrow\frac{dy}{dx}=2\Rightarrow\frac{9}{y}=2\Rightarrow y=\frac{9}{2}$

Putting in $y^2=18x\Rightarrow x=\frac{9}{8}$

$\therefore$ Required point is $\left(\frac{9}{8},\frac{9}{2}\right)$

67. Given a cone with angle $\tan^{-1}\frac{3}{4}$

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$\tan\theta = \frac{3}{4} = \frac{r}{h}$

$\frac{dV}{dt} = 6$

$V = \frac{1}{3}\pi r^2 h = \frac{1}{3}\pi h^3 \tan^2\theta = \frac{9\pi}{48}h^3 = \frac{3\pi}{16}h^3$

$\Rightarrow \frac{dv}{dt} = \frac{3\pi}{16}.3h^2.\frac{dh}{dt} = 6 \Rightarrow \left(\frac{dh}{dt}\right)_{h=4} = \frac{2}{3\pi} m/hr$ ....(i)

Now, $S = \pi rl = \frac{15}{16}\pi h^2$

Differentiate w.r.t. 't' both sides.

$\Rightarrow \frac{dS}{dt} = \frac{15\pi}{16}.2h\frac{dh}{dt}$ ....{from (i)}

$\Rightarrow \left(\frac{dS}{dt}\right)_{h=4} = 5m^2/hr$

68. Since, $g(x) = 3f\left(\frac{x}{3}\right) + f(3-x)$ and $f''(x) > 0 \forall x \in (0,3)$

$\Rightarrow f'(x)$ is increasing function.

$g'(x) = 3 \times \frac{1}{3}.f'\left(\frac{x}{3}\right) - f'(3-x) = f'\left(\frac{x}{3}\right) - f'(3-x)$

If g is decreasing in $(0,\alpha) \Rightarrow g'(x) < 0$

$f'\left(\frac{x}{3}\right) - f'(3-x) < 0 \Rightarrow f'\left(\frac{x}{3}\right) > f'(3-x)$

$\Rightarrow \frac{x}{3} < x - x \Rightarrow x < \frac{9}{4}$

Therefore $\alpha = \frac{9}{4}$. Then $8\alpha = 8 \times \frac{9}{4} = 18$

69. Given that

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$f(x)=3\sin^4 x+10\sin^3 x+6\sin^2 x-3, x\in\left[-\frac{\pi}{6},\frac{\pi}{2}\right]$

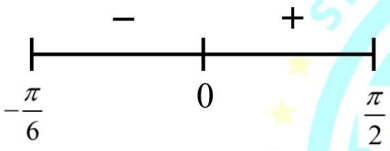
$f'(x)=12\sin^3 x\cos x+30\sin^2 x\cos x+12\sin x\cos x$

$=6\sin x\cos x(2\sin^2 x+5\sin x+2)$

$=6\sin x\cos x(2\sin x+1)(\sin x+2)=0$

$\Rightarrow x=-\frac{\pi}{6},0,\frac{\pi}{2}$

Decreasing in $\left(-\frac{\pi}{6},0,\frac{\pi}{2}\right)$



70. Here surface area of a cuboid is $=2(lb+bh+lh)$ and surface area of a closed surface area of a closed hemisphere is $3\pi r^2$ Surface area $76x^2+3\pi r^2=$ constant (K)

$r^2=\frac{K-76x^2}{3\pi}$ $\quad\left[\therefore 76x^3+3\pi r^2=K\right]$

$r=\left(\frac{K-76x^2}{3\pi}\right)^{\frac{1}{2}}$

Volume of cuboid is $l.b.h$ and

Volume of hemisphere is $\frac{2}{3}\pi r^3$

$V=40x^3+\frac{2}{3}\pi r^3; V=40x^3+\frac{2}{3}\pi\left(\frac{K-76x^2}{3\pi}\right)^{\frac{3}{2}}$

$\frac{dV}{dx}=120x^2+\frac{2}{3}\pi.\frac{3}{2}\left(\frac{K-76x^2}{3\pi}\right)^{\frac{1}{2}}\left(\frac{-76(2x)}{3\pi}\right)$

Put

$\frac{dV}{dx}=0\Rightarrow 120x^2+\frac{2}{3}\pi.\frac{3}{2}\left(\frac{K-76x^2}{3\pi}\right)^{\frac{1}{2}}.\left(\frac{-76(2x)}{3\pi}\right)=0$

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$$\Rightarrow 120x^2 = \frac{152x}{3}\left(\frac{k-76x^2}{3\pi}\right)^{\frac{1}{2}}$$

$$\Rightarrow \frac{45}{19}x^2 = x\left(\frac{k-76x^2}{3\pi}\right)^{\frac{1}{2}};\ x \neq 0$$

$$\Rightarrow \frac{45}{19}x = \left(\frac{k-76x^2}{3\pi}\right)^{\frac{1}{2}} \Rightarrow \left(\frac{45}{19}\right)^2 x^2 = \frac{k-76x^2}{3\pi}$$

$$\Rightarrow \left(\frac{45}{19}\right)x^2 = r^2 \Rightarrow \frac{x^2}{r^2} = \left(\frac{19}{45}\right)^2 \Rightarrow \frac{x}{r} = \frac{19}{45}$$

71. Given function is $f(x) = |x^2 - 5x + 6| - 3x + 2$

$$f(x) = \begin{cases} x^2 - 8x + 8 & ;\ x \in [-1,2] \\ -x^2 + 3x - 4 & ;\ x \in [2,3] \end{cases}$$

Check the values of function at $x = -1, 2, 3$.

$f(-1) = 17, f(2) = -4, f(3) = -7$

So, absolute Maximum value =17 and absolute minimum value = -7

Required sum $= 17 - 7 = 10$

72. Let $\overline{\beta}_1 = \lambda\overline{a}, \overline{\beta}_2 = \overline{\beta} - \overline{\beta}_1$

$= \left(\hat{i} + 2\hat{j} - 4\hat{k}\right) - \lambda\left(4\hat{i} + 3\hat{j} + 5\hat{k}\right)$

$= (1 - 4\lambda)\hat{i} + (2 - 3j)\hat{j} - (5\lambda + 4)\hat{k}$ and $\vec{\beta}_2.\vec{\alpha} = 0$

$\Rightarrow 4(1 - 4\lambda) + 3(2 - 3\lambda) - 5(5\lambda + 4) = 0$

$\Rightarrow 4 - 16a + 6 - 9\lambda - 25\lambda - 20 = 0 \Rightarrow 50\lambda = -10$

$\Rightarrow \lambda = \frac{-1}{5}$ Now, $\vec{\beta}_2 = \left(1 + \frac{4}{5}\right)\hat{i} + \left(2 + \frac{3}{5}\right)\hat{j} - (-1 + 4)\hat{k}$

$\vec{\beta}_2 = \frac{9}{5}\hat{i} + \frac{13}{5}\hat{j} - 3\hat{k} \Rightarrow 5\vec{\beta}_2 = 9\hat{i} + 13\hat{j} - 15\hat{k}$

So, $5\vec{\beta}_2.\left(\hat{i} + \hat{j} + \hat{k}\right) = 9 + 13 - 15 = 7$

73. Given that

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$\vec{r}\times\vec{b}=\vec{c}\times\vec{b}\Rightarrow\left(\vec{r}-\vec{c}\right)\times\vec{b}=0\Rightarrow\vec{r}-\vec{c}=\lambda\vec{b}$

$\Rightarrow\vec{r}=\vec{c}+\lambda\vec{b}$ Now, $\vec{r}.\vec{a}=0\Rightarrow\left(\vec{c}+\lambda\vec{b}\right).\vec{a}=0$

$\Rightarrow\vec{c}.\vec{a}+\lambda\vec{b}.\vec{a}=0\Rightarrow\lambda=-\dfrac{\vec{c}.\vec{a}}{\vec{b}.\vec{a}}=-\dfrac{(5-6+9)}{(1-2+6)}=-\dfrac{-8}{5}$

$r=\dfrac{5\left(5\hat{i}-3\hat{i}+3\hat{k}\right)-8\left(\hat{i}-\hat{j}+2\hat{k}\right)}{5}\Rightarrow\vec{r}=\dfrac{17\hat{i}-17\hat{j}+\hat{k}}{5}$

$\left|\vec{r}\right|=\dfrac{1}{25}(289+50)\Rightarrow25\left|\vec{r}\right|^2=339$

74. Given equations are

$l+m-n=0\Rightarrow n=l+m$

Now, put the value of n in another equation then,

$3l^2+m^2+cl(l+m)=0$

$3l^2+m^2+cl^2+clm=0$

$(3+c)l^2+clm+m^2=0$

$(3+c)\left(\dfrac{l}{m}\right)^2+c\left(\dfrac{l}{m}\right)+1=0$ …(i)

Given that lines are parallel.

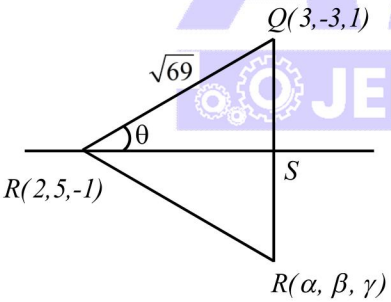
Then, roots of (i) must be equal $\Rightarrow D=0$

$c^2-4(3+c)=0\Rightarrow c^2-4c-12=0$

$(c-6)(c+2)=0\Rightarrow c=6$ or $c=-2$

Therefore, $+ve$ value of $c=6$

75.



$RQ=\sqrt{1+64+4}=\sqrt{69}$

$\overrightarrow{RQ}=\hat{i}-8\hat{j}+2\hat{k}$

Vector along $\overrightarrow{RS}=\hat{i}+\hat{j}-\hat{k}$

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$$\cos\theta = \frac{\overrightarrow{RQ}.\overrightarrow{RS}}{\left|\overrightarrow{RQ}\right|\left|\overrightarrow{RS}\right|} = \left|\frac{1-8-2}{\sqrt{69}\sqrt{3}}\right| = \frac{9}{3\sqrt{23}}$$

$$\cos\theta = \frac{3}{\sqrt{23}}$$

$$RS = \sqrt{69} \times \frac{3}{23} = 3\sqrt{3}$$

$$\sin\theta = \frac{\sqrt{14}}{\sqrt{23}} = \frac{QS}{\sqrt{69}}$$

$$QS = \sqrt{42}$$

$$\text{Area} = \frac{1}{2}.2QS.RS = \sqrt{42}.3\sqrt{3}$$

$$\lambda = 9\sqrt{14}$$

$$\lambda^2 = 81.14 = 14k$$

$$k = 81$$

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